POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2021
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Do you consider *Text to Speech applications* such as Siri and Alexa as an intelligent agent? Explain with appropriate examples. 7
   b) What do you mean by Total Turing Test? Explain in detail. 8
2. a) Your task is to get a robot out of a maze. The robot can turn to face East, West, North or South. Formulate this problem. 7
   b) Explain with an example how A-star search can find the optimal solution when there is more than one solution. 8
3. a) What is minimax search for game playing? Explain the minmax algorithm with suitable example. 7
   b) Represent the following using first order predication logic. 8
      i) Anyone whom Sophie loves is a cricket star.
      ii) Any student who does not pass does not play.
      iii) Paras is a student.
      iv) Any student who does not study does not pass.
      v) Anyone who does not play is not a cricket star.
      vi) Everyone loves Abhi.
      vii) People protest the politicians they dislike.
      viii) Not all staffs are loyal to their employers.
4. a) Explain Bayesian networks with example. 8
   b) Define and explain supervised and unsupervised learning. Describe briefly the applications of reinforcement learning? 7

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5. a) What is Clustering? Explain KNN algorithm with an appropriate example. 8

   b) Explain the explanation based learning with an example. 7

6. a) Define the role of bias in a perceptron. How are weights updated in a single layer perceptron? Explain with example. 7

   b) How can ambiguity occur at phonetic, syntactic, semantic and pragmatic levels of natural language processing? Explain with examples. 8

7. Write short notes on: (**Any two**) 2×5

   a) Constraint Satisfaction Problem

   b) Expert System

   c) Machine Vision

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